

```
[root@vbg services]# setsebool httpd_enable_cgi off
```

You can check the new value of the Boolean by again using the *getsebool* command described above:

```
[root@vbg services]# getsebool httpd_enable_cgi
httpd_enable_cgi --> off
```

The above change will affect the Boolean value in the currently loaded policy but will not remain after reboot. To make Boolean values persistent across reboots, use the *-P* option with the *setsebool* command:

```
[root@vbg services]# setsebool -P httpd_enable_cgi off
```

This will ensure that value of the *httpd_enable_cgi* Boolean has been set to *off* and will not change even after reboot.

```
[root@vbg services]# togglesetbool httpd_enable_cgi
httpd_enable_cgi: active
[root@vbg services]# getsebool httpd_enable_cgi
httpd_enable_cgi --> on
```

A note of caution though—*togglesetbool* only changes the “in memory” value of a Boolean. Changes made using the *togglesetbool* command are not persistent across reboots.

As an exercise, I leave it to you to discover the Boolean that disables SELinux policy rules from applying to a particular service or daemon. In case of doubt, you can mail me for the answer.

Booleans also help to understand the various protected daemons under the SELinux Targeted Policy.

PortCon

PortCon or Port Security Contexts are similar to File Security Contexts, but are applied to Network Sockets or ports. In a SELinux policy, access to various ports by subjects is critical. Portcon or Port Security Context Labels are based on protocol and port number(s)/range.

As an example, under the SELinux Targeted Policy, the *httpd* subject can only listen on standard ports. If the system administrators change the default ports from 80(http) or 443(https), they will need to add a new portcon in the SELinux policy to allow *httpd* to bind to this port.

To view the defined Port Security Contexts in the loaded SELinux policy, issue the following command:

```
[root@vbg ~]# seinfo -p
```

As you can see, the Port Context information contains the protocol, port and the security context. As an example, the *http_port* security contexts defined in the Red Hat Targeted policy are:

```
portcon tcp 80 system_u:object_r:http_port_t:s0
portcon tcp 443 system_u:object_r:http_port_t:s0
portcon tcp 488 system_u:object_r:http_port_t:s0
portcon tcp 8008 system_u:object_r:http_port_t:s0
portcon tcp 8009 system_u:object_r:http_port_t:s0
portcon tcp 8443 system_u:object_r:http_port_t:s0
```

As you can see, the first line of the output above labels the network Port 80 for TCP protocol with the security context:

```
system_u:object_r:http_port_t:s0
```

For the GUI initiated, policy details such as Booleans and Port Contexts can be viewed/modified by the *system-config-selinux* command:

```
[root@vbg services]# system-config-selinux
```

Do execute this command under the X environment to view and comprehend the building blocks of a SELinux Security Policy.

NodeCon

Node Security contexts are labelled by the nodecon statements in an SELinux policy. They can be used to apply access control restrictions from various hosts/nodes in the network. An effective security policy for network access to services can be created by creatively applying node context access restrictions.

To list the default node contexts in the loaded SELinux policy, issue the following command:

```
[root@vbg ~]# seinfo -o
```

The output specifies security contexts assigned to various hosts:

```
nodecon 0.0.0.0 255.255.255.255 system_u:object_r:inaddr_any_node_t:s0
nodecon 127.0.0.1 255.255.255.255 system_u:object_r:lo_node_t:s0
nodecon :ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff system_u:object_r:unspec_node_t:s0
nodecon :ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff system_u:object_r:compat_ipv4_node_t:s0
nodecon :ffff:0000:0000:ffff:ffff:ffff:ffff:ffff system_u:object_r:mapped_ipv4_node_t:s0
nodecon fe80::ffff:ffff:ffff:: system_u:object_r:link_local_node_t:s0
nodecon fe00::ff00:: system_u:object_r:site_local_node_t:s0
nodecon ff00::ff00:: system_u:object_r:multicast_node_t:s0
```

Classes

Classes, or more specifically ‘Object Classes’, are the resources on which SELinux Access Restrictions are applied. Examples of Object Classes include file, dir and network sockets. An instance of an Object Class is called an Object.

To see the various Object Classes in the default loaded SELinux Policy, issue the following command: